Lecture 32: Unsupervised Machine Learning (Part-2) — Dimensionality Reduction & Anomaly Detection

Welcome to **Lecture 32: Unsupervised Machine Learning (Part-2)**. This lecture comprehensively covers two foundational pillars of modern unsupervised learning:

1. **Dimensionality Reduction**: Principal Component Analysis (PCA), Curse of Dimensionality, Hughes Phenomenon, and geometric eigen-decomposition.
2. **Anomaly & Outlier Detection**: Density-based, tree-based, and neighborhood-relative algorithms: DBSCAN, Isolation Forest, and Local Outlier Factor (LOF).

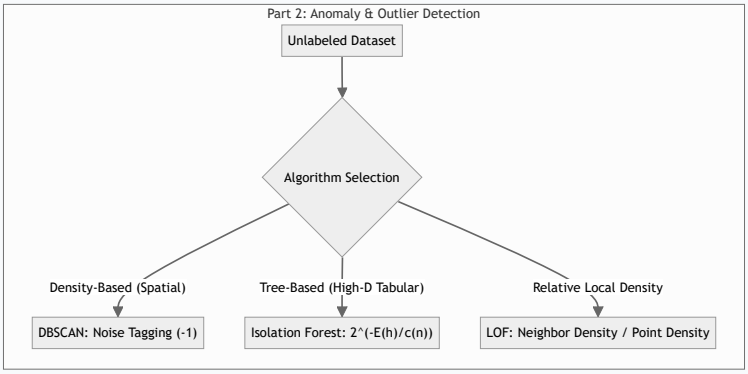
Complete Curriculum & Submodule Navigation

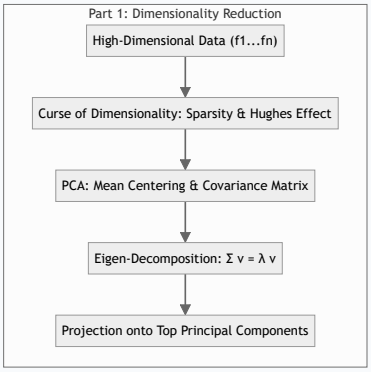
Submodule	Topic & Title	Type	Code Artifacts	Description
`Notes_32.01`	**Dimensionality Reduction & Curse of Dimensionality**	Theory	—	Data sparsity ($O(k^d)$), Euclidean distance failure, computational bottlenecks, Hughes phenomenon, and feature selection vs extraction.
`Notes_32.02`	**What is PCA - Intuition & Principles**	Theory	—	Netflix user engagement intuition, correlation requirement, data spread $\equiv$ variance, mean centering, PC1 & orthogonal PC2.
`Notes_32.03`	**PCA Step-by-Step Mathematical Algorithm**	Theory	—	Z-score standardization, covariance matrix Σ, Pythagorean equivalence ($\min d_1 \iff \max d_2$), $\Sigma\mathbf{v} = \lambda\mathbf{v}$, and projection matrix W.
`Notes_32.04`	**PCA Implementation on Iris Dataset**	Practical	`pca.ipynb`	Scikit-Learn `PCA(n_components=2)`, explained variance ratio (72.96% & 22.85%), eigenvector loadings, and 2D scatter visualization.
`Notes_32.05`	**What is Anomaly Detection -**	Theory	—	Fundamental distinction: Outliers (extreme but valid) vs Anomalies (suspicious

Submodule	Topic & Title	Type	Code Artifacts	Description
	Outlier vs Anomaly			violations); real-world domains: finance, QC, healthcare, cyber, bots.
`Notes_32.06`	**DBSCAN for Anomaly Detection**	Theory	—	Density-based noise isolation, core vs border vs noise (`label = -1`), comparison against K-Means forced assignment.
`Notes_32.07`	**DBSCAN Implementation for Anomaly Detection**	Practical	`clustering_DBSCAN.ipynb`	Non-linear `make_moons` clustering, hyperparameter heuristics ($\text{min_samples} \approx 2d = 5, \epsilon = 0.18$), and scatter plot with detected noise.
`Notes_32.08`	**Isolation Forest for Anomaly Detection**	Theory	—	Isolation trees (iTrees), random orthogonal partitioning, path length $h(x)$, anomaly scoring formula $s(x, n) = 2^{-E(h(x))/c(n)}$, and BST normalization.
`Notes_32.09`	**Isolation Forest on Clinical Thyroid Dataset**	Practical	`isolation_forest.ipynb` `thyroid_dataset.csv`	Clinical screening of 6,916 patient records across 21 features with `contamination=0.036` , detecting 249 hypothyroid pathology outliers.
`Notes_32.10`	**Local Outlier Factor (LOF) for Anomaly Detection**	Theory	—	Multi-density cluster breakdown, local vs global outliers, k-distance, reachability distance, local reachability density (LRD), and LOF ratio scoring.
`Notes_32.11`	**Local Outlier Factor Implementation**	Practical	`local_outlier_factor.ipynb` `thyroid_dataset.csv`	End-to-end LOF implementation on 21-feature clinical data detecting 249 local outliers with 2D PCA visual projections.

Submodule	Topic & Title	Type	Code Artifacts	Description
`Practice_Problems`	**Practice Case Studies & Solutions**	Problems & Solutions	`Solutions.md`	3 rigorous real-world case studies: E-Commerce PCA analytics, semiconductor wafer defect detection, and clinical endocrinology diagnostics.
`Lecture_Code`	**Master Official Code Directory**	Code Directory	All Notebooks & CSV	Master archive containing official unmodified instructor notebooks and datasets.

Core Algorithmic Architecture





Architectural Rules Followed

- **No Synthetic Code in Theory Submodules**: Pure theory modules (`32.01` , `32.02` , `32.03` , `32.05` , `32.06` , `32.08` , `32.10`) contain exclusively rich mathematical expositions, blackboard mappings, and diagrams with zero artificial notebooks.
- **Official Code Fidelity**: Code submodules (`32.04` , `32.07` , `32.09` , `32.11`) contain clean, tested Jupyter notebooks matching the exact code taught in lecture.
- **MathJax Column 0 Alignment**: All display math expressions are unindented at column 0 for universal rendering across GitHub markdown, Pandoc, and PDF converters.
- **Bilingual Dual Deliverables**: Every module provides both an English (`README.md`) and Hinglish (`README_HINGLISH.md`) guide, accompanied by compiled vector-grade PDFs.